User Acceptance Testing (UAT) Template

Date	19 February 2026
Team ID	LTVIP2026TMIDS83753
Project Name	Weather-Based Prediction of Wind Turbine Energy Output: A Next-Generation Approach to Renewable Energy Management

Project Overview:

Project Name: Weather-Based Prediction of Wind Turbine Energy Output

Project Description:
This project is a machine learning-based web application that predicts wind turbine energy output using weather parameters such as wind speed and theoretical power. The system uses a trained Random Forest regression model to generate real-time predictions through a Flask-based dashboard interface.

Project Version: 1.0

Testing Period: 16 February 2026 to 19 February 2026

Testing Scope:

Features & Functionalities Tested:

- User Input (Wind Speed & Theoretical Power)Image Format Validation
- Input Validation (Numeric & Required Fields)Confidence Score Display
- Model Prediction (Energy Output)Dashboard Visualizations
- Real-Time Output Display
- Error Handling (Missing or Invalid Input)
- Dashboard UI Display
- Prediction Accuracy Verification

User Stories Tested:

- User can enter wind speed and theoretical power values.
- User can click “Predict Energy” to generate output.
- User receives predicted energy output instantly.
- User receives error message if invalid input is entered.
- User can modify inputs and re-run prediction.

Testing Environment:

URL/Location: Local Web Application (Flask – 127.0.0.1:5000)
Credentials (if required): Not Required

Test Cases:

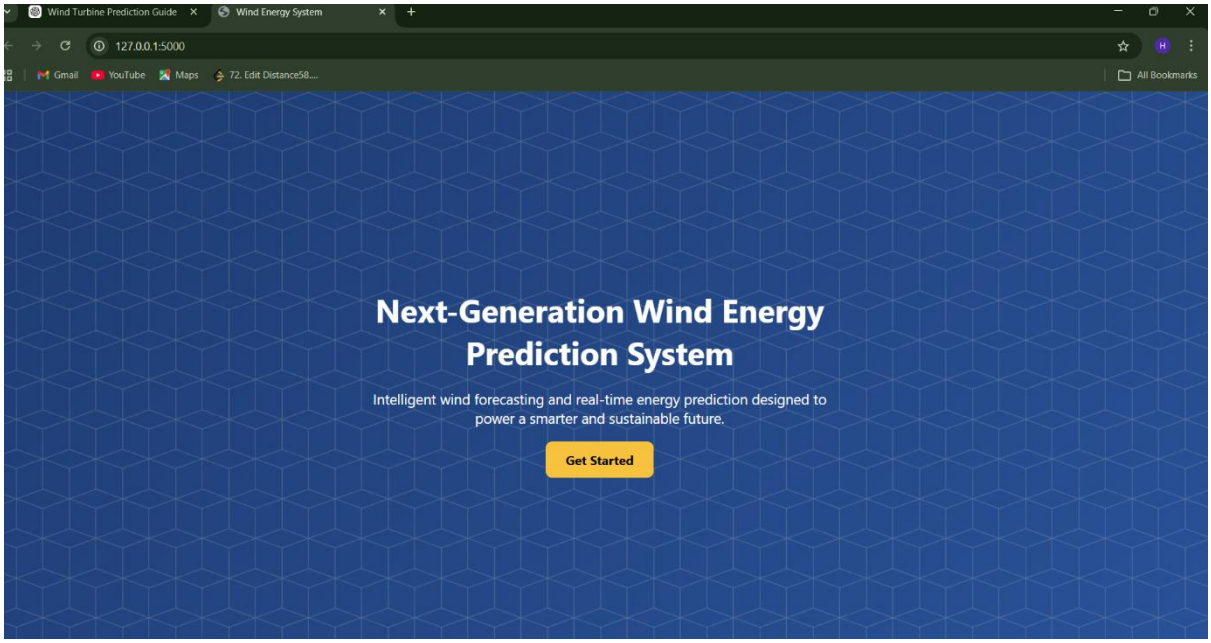
Test Case ID	Test Scenario	Test Steps	Expected Result	Actual Result	Pass/Fail
TC-001	Enter valid input values	Step 1: Enter wind speed Step 2: Enter theoretical power Step 3: Click Predict	Predicted energy output displayed successfully	Prediction displayed correctly.	Pass
TC-002	Enter non-numeric input	Step 1: Enter text instead of number Step 2: Click Predict.	System should show validation error	Error message displayed.	Pass
TC-003	Missing input fields	Step 1: Leave wind speed empty Step 2: Click Predict	System should prompt required field message	Validation message displayed	Pass
TC-004	Check prediction accuracy	Step 1: Enter sample dataset values Step 2: Click Predict	Output should be within expected range	Output within acceptable range	Pass
TC-005	Multiple predictions	Step 1: Enter new values Step 2: Click Predict again	System recalculates and shows updated result	Updated result displayed	Pass
TC-006	Page refresh	Step 1: Refresh page Step 2: Enter values again	System should function normally	Working correctly normally	Pass

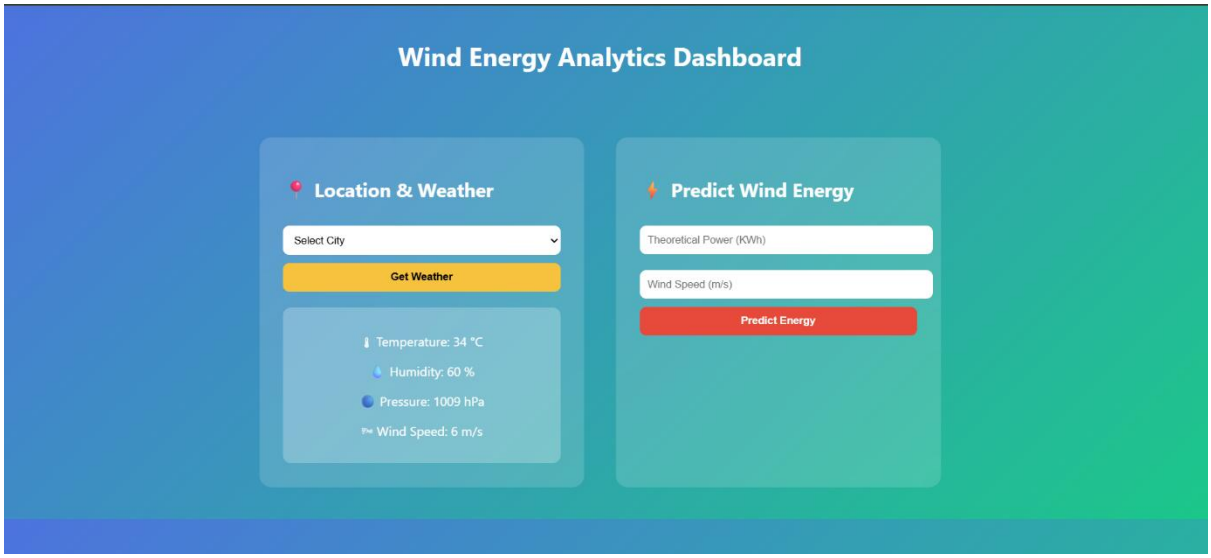
Bug Tracking:

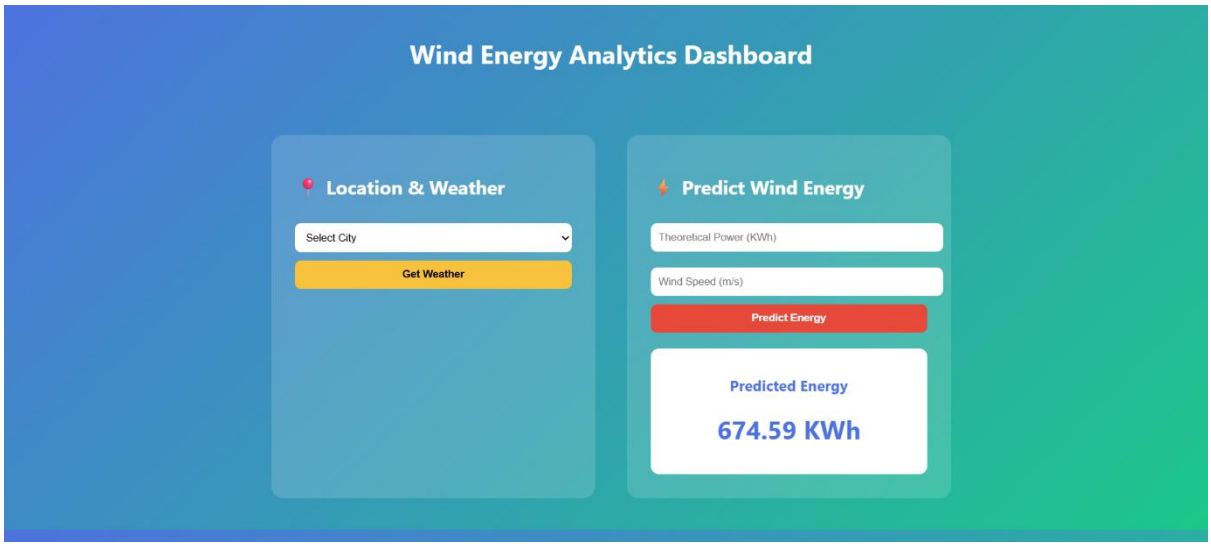
Bug ID	Bug Description	Steps to reproduce	Severity	Status	Additional feedback
BG-001	Incorrect prediction for extreme wind speed values	Step 1: Enter very high wind speed Step 2: Click Predict	Medium	Closed	Model retrained with extended dataset
BG-002	Slow response during first prediction.	Step 1: Start application Step 2: Click Predict	Low	Closed	Model preloaded to reduce delay
BG-003	Improper validation for negative values	Step 1: Enter negative wind speed Step 2: Click Predict	Medium	Closed	Added input validation rules

Outputs:

UI screenshots (Home,Dashboard).




Wind Energy Analytics Dashboard
Location & Weather
Select City
Get Weather
Temperature: 34 °C
Humidity: 60 %
Pressure: 1009 hPa
Wind Speed: 6 m/s
Predict Wind Energy
Theoretical Power (KWh)
Wind Speed (m/s)
Predict Energy


Wind Energy Analytics Dashboard
Location & Weather
Select City
Get Weather
Predict Wind Energy
Theoretical Power (KWh)
Wind Speed (m/s)
Predict Energy
Predicted Energy
674.59 KWh